

Homological Flat Bands and Rank-Cubic Spectral Geometry in SU(N) Lattice Gauge Theory

An expanded theorem program built from one-plaquette Weyl analysis and strong-coupling plaquette dynamics

Prepared from the supplied SU(N)/Weyl-gap manuscript suite

July 23, 2026

Contents

- **Core theorem:** Sections 2–5 derive the corrected bridge, exact $SU(N)$ channel coefficient, homological band spectrum, and N^{-3} mobility.
- **Special-rank and local theory:** Sections 6–8 recover the $SU(3)$ result, state the source-certified third-order extension, and organize the Weyl-local gap theorems.
- **Research program:** Sections 9–13 define the Weyl–Hodge synthesis, status ledger, publication architecture, and verification gates.
- **Audit trail:** Appendices A–C record the exact algebra, certificate coverage, and source provenance.

Abstract

This manuscript develops a larger theory from two bodies of supplied work: the compact one-plaquette class Hamiltonian at large potential strength and the strong-coupling one-plaquette glueball band on the three-dimensional cubic lattice. Its new proved core is a fixed-rank $SU(N)$ extension of the charge-odd flat-band mechanism. At second order in the magnetic coupling, the orientation-sensitive hopping coefficient is

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}, \quad N \geq 3.$$

It is positive, reduces to $5/612$ at $N = 3$, and obeys $N^3 t_N \rightarrow 1/4$. The spatial matrix multiplying t_N is the signed face-adjacency operator

$$S(\mathbf{k}) + 4I = \tilde{N}(\mathbf{k})\tilde{N}(\mathbf{k})^\dagger,$$

where $\tilde{N}$ is the Bloch face-to-edge boundary matrix. Because $\partial_2 \partial_3 = 0$, $S(\mathbf{k})$ has an exactly flat eigenvalue -4 for every momentum. In fact its full spectrum is

$$\text{spec} S(\mathbf{k}) = \{-4, q(\mathbf{k}) - 4, q(\mathbf{k}) - 4\}, \quad q(\mathbf{k}) = 4 \sum_{j=1}^3 \sin^2 \frac{k_j}{2}.$$

Consequently the lowest C -odd branch is dispersionless through $O(y^2)$ for every $SU(N \geq 3)$, the full one-plaquette C -odd manifold has width

$$W_N^{(-)}(y) = 12t_N y^2 + O(y^3) = \frac{24N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} y^2 + O(y^3),$$

and $W_N^{(-)} \sim 3y^2/N^3$ at fixed strong-coupling normalization. On an L^3 torus the flat eigenspace has exact dimension $L^3 + 2$.

The second half of the manuscript organizes the supplied local spectral results into a “Weyl–Hodge” program. The internal axis is controlled by Weyl-invariant degree and yields the exact $SU(3)$ three-term local gap. The spatial axis is controlled by the cellular boundary complex and yields the homological flat band. A second rank-cubic scale appears on the local side: the fixed- N , large- β class-gap expansions are uniform only in the regime $\beta \gg N^3$. The common exponent three is a structural clue, not yet a theorem connecting the two asymptotic regimes.

The manuscript distinguishes four statuses throughout: proved here; inherited exact result from the supplied derivations; source-certified but not independently rerun; and conjectural/open. It makes no claim about the continuum Yang–Mills mass gap.

Reader’s guide and status vocabulary

The goal is not to concatenate the supplied documents. It is to extract one stronger paper and one credible research program.

The status labels used below are:

Label	Meaning
Proved here	A complete algebraic or geometric derivation appears in this manuscript and is checked by the accompanying exact-arithmetic certificate.
Inherited exact	The result is proved in the supplied manuscript suite and its formulas agree across the relevant source files.

Label	Meaning
Source-certified	The supplied derivation names exact-arithmetic certificates, but those certificate files were not among the attachments and were not rerun here.
Conjectural/open	The statement is a proposed scaling law, structural interpretation, or unresolved analytic target.

Main result. Under the standard second-order linked-cluster reduction of the one-plaquette $\mathcal{C}$ -odd manifold, the signed nearest-neighbor coefficient is the exact positive rational function t_N displayed above. The signed adjacency is a shifted boundary Laplacian, so the lowest branch is flat for every $SU(N \geq 3)$. The group dependence and the spatial topology factor completely at this order.

The paper's central factorization is therefore

$$\text{second-order } \mathcal{C}\text{-odd dynamics} = \text{rank coefficient } t_N \times \text{cubical boundary geometry } S$$

up to a momentum-independent scalar.

1. Scope and result hierarchy

The supplied work naturally separates into three layers.

1. **Local compact spectral theory.** The class Hamiltonian

$$H_\beta = \frac{1}{2}C_2 + \beta \left(1 - \frac{1}{N} \text{ReTr}g\right)$$

is analyzed at fixed rank as $\beta \rightarrow \infty$. For $SU(3)$, Weyl geometry gives a three-term local class gap with an exact nonradial correction.

2. **Strong-coupling spatial dynamics.** A one-plaquette excitation of the Kogut–Susskind Hamiltonian hops among plaquettes. In the C -odd sector the hop carries the orientation sign of the shared edge. The corresponding Bloch operator is a shifted face-boundary Laplacian and has a homological flat band.
3. **Global Wilson geometry.** The supplied Peierls and projected-capacity documents propose a route from local defects to Wilson-loop control. That route remains conditional. It is not upgraded merely by strengthening the local spectral layer.

This manuscript enlarges layers 1 and 2 and keeps layer 3 separate. The strongest publishable unit is the rank-universal flat-band theorem, with the $SU(3)$ local gap and third-order analysis serving as deep special-rank companions.

2. Strong-coupling Hamiltonian and the corrected bridge

2.1 Kogut–Susskind normalization

On the spatial cubic lattice, use

$$H = \sum_{\ell} \frac{1}{2} C_2(\ell) - y \sum_p (\chi_p + \bar{\chi}_p),$$

where $\chi_p = \text{Tr}U_p$ is the fundamental plaquette character and $y > 0$ is the strong-coupling expansion parameter. A single fundamental loop places F on four links, hence its unperturbed energy is

$$E_{F,N} = 4 \cdot \frac{1}{2} C_F = 2C_F = \frac{N^2 - 1}{N}, \quad C_F = \frac{N^2 - 1}{2N}.$$

For $N \geq 3$, the two oriented loops $\chi_p|0\rangle$ and $\bar{\chi}_p|0\rangle$ are distinct. Their charge eigenstates are

$$\left|e_p\right\rangle = \frac{\chi_p + \bar{\chi}_p}{\sqrt{2}}|0\rangle, \quad \left|o_p\right\rangle = \frac{\chi_p - \bar{\chi}_p}{\sqrt{2}}|0\rangle.$$

For $SU(2)$, the fundamental is pseudoreal and this distinct fundamental C -odd sector is absent. The algebraic formula below vanishes at $N = 2$, as it should.

2.2 Exact bridge theorem for every rank

Restrict the lattice Hamiltonian to the flux tower of one plaquette. A character $\chi_R(g)$ puts the representation R on all four boundary links, so the electric energy is $2C_2(R)$. The restricted operator is

$$h_{\text{loop}} = 2C_2 - 2y \text{ReTr}g.$$

On the other hand,

$$4H_\beta - 4\beta = 2C_2 - \frac{4\beta}{N} \text{ReTr}g.$$

Matching the magnetic terms gives $\beta = Ny/2$.

Theorem 2.1 — Corrected fixed-rank bridge (proved here).

$$h_{\text{loop}} = [4H_\beta - 4\beta]_{\beta=Ny/2}, \quad m_{\pm}^{\text{restricted}}(y) = 4\Delta_N^{\pm} \left(\frac{Ny}{2} \right).$$

The first equality is an exact operator identity on the one-plaquette flux tower; the second is its gap consequence.

This corrects a harmless additive-constant typo in the supplied flat-band manuscript, which wrote $4H_\beta - \beta$. The gap map used there remains correct because scalar shifts cancel in gaps, but the exact operator identity requires $4H_\beta - 4\beta$.

For $SU(3)$, the determinant tensor permits a first-order charge splitting:

$$m_+(y) = \frac{8}{3} - y + O(y^2), \quad m_-(y) = \frac{8}{3} + y + O(y^2).$$

For $N > 3$, N -ality forbids this particular first-order matrix element. The rank-universal theorem below concerns the spatial dispersion at second order and does not require a universal formula for the momentum-independent on-site coefficient.

3. Exact fixed-rank channel algebra

3.1 Why only four channels occur

Two plaquettes sharing one edge produce two possible fused representations on that edge:

$$F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj.}$$

$$F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F.$$

The relevant dimensions and Casimirs are:

Channel R	d(R)	C ₂ (R)
Singlet	1	0
Adj	N ² - 1	N
Λ ² F	N(N - 1)/2	(N + 1)(N - 2)/N
Sym ² F	N(N + 1)/2	(N - 1)(N + 2)/N

After the six nonshared links are integrated, the squared norm of channel R is d_R/N^2 . Relative to the one-plaquette energy, the two-plaquette intermediate has denominator

$$\delta E_R = C_F + \frac{1}{2} C_2(R).$$

Second-order perturbation contributes the negative inverse of this positive excitation energy.

3.2 Antiparallel channel sum

For $F \otimes \bar{F}$, define

$$A_N = - \left[\frac{1}{N^2} \frac{1}{C_F} + \frac{N^2 - 1}{N^2} \frac{1}{C_F + N/2} \right].$$

Using $C_F = (N^2 - 1)/(2N)$,

$$A_N = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}.$$

At $N = 3$, this gives $A_3 = -27/68$, the combined singlet-plus-adjoint part of the supplied $SU(3)$ channel sum.

3.3 Parallel channel sum

For $F \otimes F$, define

$$B_N = -\left[\frac{N-1}{2N} \frac{1}{C_F + \frac{1}{2}C_{A^2F}} + \frac{N+1}{2N} \frac{1}{C_F + \frac{1}{2}C_{\text{Sym}^2F}} \right].$$

The denominators factor as

$$C_F + \frac{1}{2}C_{A^2F} = \frac{(N+1)(2N-3)}{2N},$$

$$C_F + \frac{1}{2}C_{\text{Sym}^2F} = \frac{(N-1)(2N+3)}{2N}.$$

Therefore

$$B_N = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}.$$

At $N = 3$, $B_3 = -7/18$. Hence

$$A_3 + B_3 = -\frac{481}{612},$$

exactly recovering the supplied $SU(3)$ two-flux channel sum.

3.4 Charge-odd residue and common leakage

The C -odd combination subtracts the two orientation classes. Its signed nearest-neighbor coefficient is

$$t_N = B_N - A_N.$$

Putting the two rational functions over a common denominator gives

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}.$$

The numerator makes the rank dependence transparent:

$$t_2 = 0, \quad t_N > 0 \quad (N \geq 3).$$

The diagonal leakage relative to the vacuum includes the vacuum subtraction $1/C_F = 2N/(N^2 - 1)$:

$$\ell_N = A_N + B_N + \frac{1}{C_F} = \frac{2N(5 - 3N^2)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}.$$

At $N = 3$,

$$t_3 = \frac{5}{612}, \quad \ell_3 = -\frac{11}{306},$$

which are the exact hopping and per-neighbor leakage coefficients in the supplied $SU(3)$ derivation.

The algebra above is checked as a formal rational identity in N , not merely rank by rank, by the accompanying standard-library certificate.

4. The fixed-rank homological flat-band theorem

4.1 The signed face adjacency

Choose the three plaquette orbitals

$$a = (12), \quad b = (13), \quad c = (23)$$

in each cubic unit cell, with their standard oriented boundaries. Let

$$u_j = 1 - e^{ik_j}.$$

The Bloch face-to-edge boundary matrix is

$$\tilde{N}(\mathbf{k}) = \begin{pmatrix} u_2 & -u_1 & 0 \\ u_3 & 0 & -u_1 \\ 0 & u_3 & -u_2 \end{pmatrix}.$$

The signed shared-edge adjacency satisfies

$$S(\mathbf{k}) + 4I = \tilde{N}(\mathbf{k})\tilde{N}(\mathbf{k})^\dagger.$$

The shift by four is the boundary norm of one oriented plaquette. The off-diagonal terms are precisely the signed sums over shared edges.

4.2 Exact spectrum, not only one flat eigenvalue

The determinant vanishes identically:

$$\det \tilde{N}(\mathbf{k}) = 0.$$

More strongly, after row signs and a row permutation, $\tilde{N}$ is the cross-product matrix of the vector $u = (u_1, u_2, u_3)$. Hence

$$\tilde{N}^\dagger \tilde{N} \sim qI - uu^\dagger, \quad q = \sum_{j=1}^3 |u_j|^2.$$

The nonzero singular values are therefore both $\sqrt{q}$, and

$$q(\mathbf{k}) = \sum_{j=1}^3 |1 - e^{ik_j}|^2 = 4 \sum_{j=1}^3 \sin^2 \frac{k_j}{2}.$$

This gives the complete spectrum:

$$\boxed{\text{spec} S(\mathbf{k}) = \{-4, q(\mathbf{k}) - 4, q(\mathbf{k}) - 4\}}.$$

The flat eigenvector may be taken as

$$w(\mathbf{k}) = (\bar{u}_3, -\bar{u}_2, \bar{u}_1).$$

At $\mathbf{k} = 0$, all u_j vanish and the whole triplet sits at -4 .

4.3 Separation of group theory and geometry

At second order, the C -odd one-plaquette effective Hamiltonian has the form

$$H_{-,N}^{(2)}(\mathbf{k}) = a_{N,-}I + t_N S(\mathbf{k}),$$

where $a_{N,-}$ contains the within-plaquette coefficient and momentum-independent leakage. The entire momentum dependence is carried by the same signed matrix S for every rank.

Thus the three second-order coefficients are

$$a_{N,-} - 4t_N, \quad a_{N,-} + (q(\mathbf{k}) - 4)t_N \quad (\text{twice}).$$

Because $t_N > 0$ and $q \geq 0$, the flat branch is the lowest branch.

Theorem 4.1 — Fixed-rank $SU(N)$ homological flat band (proved here).

For every $N \geq 3$, the lowest fundamental C -odd one-plaquette branch of the three-dimensional cubic Kogut–Susskind Hamiltonian is exactly dispersionless through $O(y^2)$. Its two companion branches are degenerate and disperse through $q(\mathbf{k})$. The result is independent of the momentum-independent on-site coefficient.

In a full mass notation,

$$m_{-, \text{flat}}(y) = E_{F,N} + c_{N,-}^{(1)}y + (a_{N,-} - 4t_N)y^2 + O(y^3),$$

$$m_{-, \text{disp}}(\mathbf{k}, y) = E_{F,N} + c_{N,-}^{(1)}y + [a_{N,-} + (q(\mathbf{k}) - 4)t_N]y^2 + O(y^3),$$

with multiplicity two in the dispersive line. Here $c_{3,-}^{(1)} = 1$ in the supplied normalization, while the analogous first-order splitting vanishes for $N > 3$ by N -ality.

4.4 Bandwidth and curvature

Since $0 \leq q \leq 12$, the signed-adjacency spectrum ranges from -4 to 8 . The full C -odd one-plaquette manifold therefore has width

$$W_N^{(-)}(y) = 12t_N y^2 + O(y^3) = \frac{24N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} y^2 + O(y^3).$$

Near $\mathbf{k} = 0$,

$$q(\mathbf{k}) = |\mathbf{k}|^2 + O(|\mathbf{k}|^4),$$

so each dispersive companion has isotropic curvature $t_N y^2$. The flat branch has zero curvature at this order.

4.5 Position-space topology and exact degeneracy

Let

$$C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$$

be the cubical cellular chain complex. The matrix $\tilde{N}$ is the Bloch form of ∂_2 . Every elementary cube boundary is a closed two-chain:

$$\partial_2(\partial_3 \text{ cube}) = 0.$$

Therefore the six signed faces of one cube form a compactly localized eigenstate of S at eigenvalue -4 .

More generally,

$$\dim \ker \partial_2 = \text{rank } \partial_3 + b_2.$$

On the periodic L^3 cubic lattice, $b_2(T^3) = 3$ and $\text{rank } \partial_3 = L^3 - 1$. Hence

$$\dim E_{-4}(S) = L^3 + 2.$$

This is one flat state for every momentum, plus two extra states at the three-band touching point $\mathbf{k} = 0$.

5. Rank-cubic mobility

The exact coefficient has the large-rank expansion

$$t_N = \frac{1}{4N^3} + O(N^{-5}),$$

and therefore

$$W_N^{(-)}(y) = \frac{3y^2}{N^3} + O(N^{-5}y^2) + O(y^3).$$

At fixed strong-coupling normalization, the spatial mobility of the fundamental C -odd excitation is suppressed by N^{-3} , not merely N^{-1} .

N	t(N)	12t(N)	N ³ t(N)
3	5/612	5/51	0.220588
4	32/8525	384/8525	0.240235
5	5/2548	15/637	0.245290
6	128/111825	512/37275	0.247243
7	105/145112	315/36278	0.248188
8	320/658749	1280/219583	0.248714
∞	~1/(4N ³)	~3/N ³	1/4

The cancellation producing this scale is visible before asymptotics:

$$A_N \sim -\frac{1}{N}, \quad B_N \sim -\frac{1}{N}, \quad B_N - A_N \sim \frac{1}{4N^3}.$$

Thus the C -odd mobility is a subleading residue between two order- N^{-1} fusion channels.

6. The $SU(3)$ gate and third-order persistence

6.1 Complete second-order recovery

For $SU(3)$, the supplied within-plaquette C -odd coefficient is $1/2$. The twelve-neighbor diagonal leakage is

$$12\ell_3 = -\frac{22}{51},$$

so

$$a_{3,-} = \frac{1}{2} - \frac{22}{51} = \frac{7}{102}.$$

The flat shift is

$$a_{3,-} - 4t_3 = \frac{7}{102} - 4 \cdot \frac{5}{612} = \frac{11}{306},$$

and the top of the dispersive doublet is

$$a_{3,-} + 8t_3 = \frac{41}{306}.$$

Therefore

$$m_{-,flat}^{SU(3)}(y) = \frac{8}{3} + y + \frac{11}{306}y^2 + O(y^3),$$

while the full C -odd manifold width is $5y^2/51 + O(y^3)$. These are exact regressions against the supplied manuscript.

6.2 What is known at third order

The supplied flat-band derivation states that the $SU(3)$ flat branch persists through third order:

$$m_{-,flat}^{SU(3)}(y) = \frac{8}{3} + y + \frac{11}{306}y^2 - \frac{109151}{249696}y^3 + O(y^4).$$

Its mechanism has two parts:

1. every three-distinct-plaquette tromino numerator vanishes by the bare-link lemma; and
2. the remaining third-order hopping is proportional to the same signed adjacency S .

This result is **source-certified**, not independently rerun here. The manuscript names exact-arithmetic files for the domino resolvent, Haar moments, tromino classification, and regression web, but those files were not included in the attachment set. The coefficient should remain in a submission only after those certificate files are recovered and executed in a clean environment.

6.3 The first generic lifting order

The bare-link argument also identifies the first order at which a generic three-plaquette lifter can become nonzero: $O(y^4)$. The decisive test is whether the three geometric lifter classes acquire equal weights. Equality preserves the flat line; inequality produces a computable fourth-order bandwidth.

This is a sharp binary target, not an invitation to fit a dispersion numerically. The exact weight difference is the scientifically useful output in either case.

7. The internal Weyl spectral layer

The strong-coupling flat band concerns motion among spatial plaquettes. The supplied class-gap results concern localization inside the group manifold. They are different spectral problems, but both reduce complexity by exposing the correct geometry.

7.1 $SU(3)$ Cartan invariants

For the local Hamiltonian

$$H_\beta = \frac{1}{2}C_2 + \beta \left(1 - \frac{1}{3} \text{Re} \chi_{(1,0)}(g) \right),$$

choose orthonormal Cartan coordinates (x, y) and define

$$p_2 = x^2 + y^2, \quad p_3 = \frac{\sqrt{6}}{6} y(3x^2 - y^2).$$

The Weyl discriminant is

$$\Delta_W^2 = \frac{1}{2}p_2^3 - 3p_3^2.$$

Newton identities give

$$P_4 = \frac{1}{2}p_2^2, \quad P_6 = \frac{1}{4}p_2^3 + \frac{1}{3}p_3^2.$$

After the $\beta^{-1/4}$ localization scaling,

$$H_\beta = \beta^{1/2}H_0 + H_1 + \beta^{-1/2}H_2 + O(\beta^{-1}),$$

with

$$H_1 = -\frac{p_2^2}{96},$$

$$H_2 = \sqrt{6} \left(\frac{p_2^3}{11520} + \frac{p_3^2}{8640} \right).$$

The first explicitly angular Weyl-invariant term is p_3^2 , of total degree six. This is why the radial approximation survives one correction and first fails at the $\beta^{-1/2}$ coefficient.

7.2 Exact three-term $SU(3)$ local gap

The supplied invariant-basis calculation gives:

$$\Delta_{SU(3)}(\beta) = \sqrt{\frac{2\beta}{3} - \frac{5}{16} - \frac{311\sqrt{6}}{9216}\beta^{-1/2} + O(\beta^{-1})}.$$

The radial foil has coefficient

$$-\frac{327\sqrt{6}}{9216},$$

so the exact rank-two angular correction is

$$\frac{\sqrt{6}}{576}\beta^{-1/2}.$$

This shift is the direct p_3^2 matrix element. The resolvent correction from H_1 is handled in a finite invariant oscillator basis, and the supplied coefficient tables close the calculation exactly. This result is **inherited exact**.

It is a local class-function gap. It is not a glueball mass, a lattice mass gap, or a continuum Yang–Mills gap.

7.3 Fixed-rank $SU(N)$ local class gaps

The supplied fixed-rank Wick certificates give

$$\Delta_N^+(\beta) = \sqrt{\frac{2\beta}{N} - \frac{2N^2 - 3}{16N} - \frac{\sqrt{2}(6N^4 - 24N^2 + 41)}{1024N^{3/2}}} \beta^{-1/2} + O_N(\beta^{-1})$$

and

$$\Delta_N^-(\beta) = \sqrt{\frac{9\beta}{2N} - \frac{3(N^2 - 3)}{16N} - \frac{\sqrt{2}(14N^4 - 97N^2 + 290)}{1536N^{3/2}}} \beta^{-1/2} + O_N(\beta^{-1}).$$

The C -odd constant term is

$$-\frac{3(N^2 - 3)}{16N}.$$

An older alternative,

$$-\frac{3(2N^2 + 1)}{16N},$$

is superseded and should not reappear.

These fixed-rank formulas are **inherited exact**. Their remainders are not uniform in N .

8. The second rank-cubic scale

8.1 Why the fixed-rank expansion requires $\beta \gg N^3$

In either local class gap, the displayed $\beta^{-1/2}$ coefficient grows like $N^{5/2}$, while the leading oscillator gap grows like $\sqrt{\beta/N}$. Their relative size is therefore

$$\frac{N^{5/2}\beta^{-1/2}}{\sqrt{\beta/N}} \asymp \frac{N^3}{\beta}.$$

Thus the fixed-rank series is asymptotically ordered only when

$$\boxed{\beta \gg N^3}.$$

A formal 't Hooft substitution $\beta \propto N^2$ lies outside this uniformity regime and cannot be used to infer a controlled large- N local gap.

8.2 Formal double scaling at $\beta = N^3\tau$

If one nevertheless substitutes

$$\beta = N^3\tau$$

into the displayed coefficients and divides by N , the formal outer expressions become

$$\frac{\Delta_N^+(N^3\tau)}{N} \sim \sqrt{2\tau} - \frac{1}{8} - \frac{3\sqrt{2}}{512\sqrt{\tau}} + \dots,$$

$$\frac{\Delta_N^-(N^3\tau)}{N} \sim 3\sqrt{\frac{\tau}{2}} - \frac{3}{16} - \frac{7\sqrt{2}}{768\sqrt{\tau}} + \dots$$

These are **conjectural scaling targets**, not asymptotic theorems, because the N -dependence of the $O_N(\beta^{-1})$ remainder is uncontrolled.

The correct next theorem would be a uniform estimate of the form

$$\left| \frac{\Delta_N^\pm(N^3\tau)}{N} - F_\pm^{(K)}(\tau) \right| \leq \varepsilon_{N,K}(\tau),$$

with a domain $\tau \geq \tau_0 > 0$ and $\varepsilon_{N,K} \rightarrow 0$ as $N \rightarrow \infty$. No such estimate is presently established.

8.3 The rank-cubic observation

The two regimes now display the same exponent:

Regime	Rank-cubic quantity	Meaning
Strong coupling, spatial motion	$N^3 t(N) \rightarrow 1/4$	The C-odd hopping is a $1/N^3$ residue.
Large- β , internal localization	β/N^3	The fixed-rank class-gap series becomes uniformly ordered only beyond the N^3 scale.

This is the **rank-cubic observation**. It is not yet a duality, matching relation, or common limit theorem. The bridge $\beta = Ny/2$ maps $\beta \sim N^3$ to $y \sim N^2$, far outside the small- y expansion. The exponent may reflect a shared cancellation structure in low tensor moments, but that explanation remains open.

9. Weyl–Hodge spectral geometry

The enlarged program has two independent “barcodes.”

9.1 Internal Weyl barcode

The internal barcode records the first invariant degree capable of changing the gap beyond radial data.

Rank/type	First relevant invariant structure	Spectral consequence
SU(2)	only the quadratic invariant locally	genuinely radial local geometry
SU(3)	cubic invariant p_3 , entering the even potential as p_3^2 at degree six	first angular shift at $\beta^{-1/2}$
SU($N \geq 4$)	independent higher trace invariants occur earlier at the operator level	energy shifts must still be determined by oscillator selection rules

The last distinction matters: the degree at which a nonradial operator appears is not automatically the degree at which a selected gap responds. Operator barcode and gap-response barcode must be tracked separately.

9.2 Spatial Hodge barcode

The spatial barcode records the first cluster operator not reduced to the boundary complex.

Order	Spatial structure	Status
y^2	signed nearest-neighbor face adjacency $S = \partial_2 \dagger \partial_2 - 4I$ in Bloch form	flat by $\partial_2 \partial_3 = 0$
y^3 , SU(3)	bare-link selection removes three-distinct-plaquette lifters; remaining term is affine in S	source-certified flatness
y^4	dressed tromino classes can survive	first generic lifting test

This barcode is topological at second order, selection-rule protected at third order in the supplied $SU(3)$ analysis, and generically dynamical at fourth order.

9.3 A careful $SU(3)$ resonance

$SU(3)$ is special on both axes.

- Internally, the determinant constraint leaves the cubic Weyl invariant p_3 , whose square creates the first nonradial local gap shift.
- In strong coupling, the rank-three determinant tensor implies $F \otimes F \supset \overline{F}$, producing the first-order charge splitting and unusually early baryonic contractions.

Both phenomena are consequences of rank-three invariant tensor structure. They are not the same matrix element, and no equality between their coefficients is claimed. The meaningful statement is structural: the determinant tensor controls the first departure from a simpler radial or mesonic description in both local and plaquette sectors.

10. Theorem and claim ledger

Claim	Status	Evidence in this manuscript	Submission gate
Correct bridge with subtraction 4β	Proved here	direct operator matching	none
Closed channel sums $A(N)$, $B(N)$	Proved here	representation dimensions, Casimirs, exact rational reduction	exact script passes

Claim	Status	Evidence in this manuscript	Submission gate
Closed C-odd hopping $t(N)$	Proved here	$B(N) - A(N)$	independent two-plaquette audit at $N = 4, 5$ recommended
Complete signed spectrum $\{-4, q - 4, q - 4\}$	Proved here	cross-product incidence identity	exact Laurent check passes
Flat band for every $SU(N \geq 3)$ through $O(y^2)$	Proved here within the standard second-order reduction	positivity of $t(N)$, boundary factorization	independent perturbative convention audit
Torus degeneracy $L^3 + 2$	Proved here	$\text{rank}(\partial_3) + b_2$	exact $L = 2, 3$ rank gates pass
$SU(3)$ coefficient 11/306 at y^2	Inherited exact and regressed	exact recovery from $a(3, -)$ and $t(3)$	none
$SU(3)$ third-order coefficient -109151/249696	Source-certified	supplied derivation only	recover and rerun named certificates
$SU(3)$ local class gap through $\beta^{-1/2}$	Inherited exact	invariant basis and coefficient tables agree	optional independent CAS rerun
Fixed-rank $SU(N)$ local $C \pm$ gaps	Inherited exact	finite Wick certificates in source suite	preserve corrected odd constant
Uniform $\beta = N^3 \tau$ local limit	Conjectural/open	formal coefficient matching only	prove a rank-uniform remainder
$O(y^4)$ survival of flatness	Open	equal-lifter criterion identified	compute exact fourth-order weights
Global Wilson-area or mass-gap conclusion	Not established	local results do not supply global polymer control	separate global theorem required

11. What the larger paper should be

11.1 Submission-sized core

The most defensible first paper is:

Homological C -odd Flat Bands in Strong-Coupling $SU(N)$ Lattice Gauge Theory

Its core sections would be:

1. fixed-rank bridge and perturbative normalization;
2. exact channel sums A_N, B_N ;
3. the closed hopping formula t_N ;
4. incidence factorization and complete Bloch spectrum;
5. compact cube states and torus degeneracy;
6. N^{-3} mobility;
7. $SU(3)$ second-order regression;
8. optional $SU(3)$ third-order section after certificate recovery.

This paper has a single message: **representation theory supplies one scalar, homology supplies the band.**

11.2 Companion local-spectral paper

The local class-gap material supports a separate paper:

Weyl-Invariant Perturbation Theory for Fixed-Rank $SU(N)$ One-Plaquette Gaps

Its strongest theorem is the exact $SU(3)$ three-term gap, followed by the fixed-rank $C^\pm$ Wick certificates and the explicit nonuniformity theorem. The rank-cubic double-scaling problem belongs in its outlook, not its theorem statement.

11.3 Why the global Wilson program should remain separate

The current projected-capacity layer does not create a stronger local theorem. Under the normalization used in the supplied source suite, the projected multiplier already satisfies the contraction bound

$$\|P\chi_r P\| \leq 1.$$

The genuinely difficult global inputs are free-energy subtraction, connected polymer control, and conversion of local class penalties into contour weights. Those are not consequences of either the local class gap or the homological flat band.

Combining all three layers into one paper would weaken the logical message. The local and spatial results can be cross-referenced, but the global Wilson chain should remain a clearly conditional sequel.

12. Verification and falsification program

Before submission, the following gates should be run in order.

Gate A — fixed-rank channel audit

1. Construct the exact two-plaquette character basis at $N = 4$ and $N = 5$.
2. Compute the second-order C -odd hopping directly.
3. Verify $t_4 = 32/8525$ and $t_5 = 5/2548$.
4. Check that no rank-specific epsilon block adds a second spatial matrix at this order.

The attached certificate proves the rational channel algebra. Gate A checks that the perturbative identification of those channel sums with the C -odd hopping has no hidden convention mismatch.

Gate B — finite-lattice geometry

1. Build ∂_2 exactly on $L = 2, 3, 4$ periodic cubic lattices.
2. Verify $S = \partial_2^\top \partial_2 - 4I$ in the chosen face convention.
3. Confirm $\dim E_{-4} = L^3 + 2$.
4. Fourier-transform and compare against $\{-4, q - 4, q - 4\}$.

The accompanying certificate completes this gate for the incidence identities and exact ranks at $L = 2, 3$.

Gate C — recover the $SU(3)$ third-order certificates

The supplied manuscript names:

- glueball_band_certificate.py;
- tromino_contract_independent_check.py;
- tromino_weight_constraint_certificate.py;
- su3_moments_ext.py;
- su3_domino_d3.py;
- master_v2_regression_certificate.py.

They should be recovered, hashed, and rerun in a clean environment. Until then, the third-order coefficient is not part of the independently audited core.

Gate D — fourth-order binary test

Compute the three activated lifter weights

$$W(\text{path}^+), \quad W(\text{path}^-), \quad W(\text{corner}).$$

Test equality before inspecting any fitted band. If equal, seek a structural identity. If unequal, contract the exact differences with the known geometric class matrices to obtain the first nonzero flat-band width.

Gate E — rank-uniform local gap

Bound the omitted local terms in the scaled operator uniformly when $\beta = N^3\tau$. A successful theorem must specify:

- the τ -domain;
- the rank dependence of the resolvent bounds;
- the number of retained invariant oscillator sectors;
- the norm in which the remainder is controlled.

13. Claims explicitly excluded

The enlarged theory does **not** establish any of the following:

- a continuum Yang–Mills mass gap;
- a continuum glueball mass;
- a Wilson area law from the local class gap alone;
- a uniform 't Hooft-limit expansion obtained by substituting $\beta \propto N^2$;
- all-order flatness of the C -odd band;
- a theorem equating the internal cubic invariant with the spatial cube-boundary mode;
- a formal priority or novelty claim against the full literature.

The statements proved here are fixed-lattice, fixed-order, and normalization-specific. Their value is that they are exact within that scope.

14. Positioning relative to current work

The compact one-plaquette problem sits within a long line of matrix-model and integrable-system work. Finite- N one-plaquette unitary matrix models have been studied explicitly, and the A_n trigonometric Calogero–Sutherland

system provides a broader integrable background for Weyl-symmetric spectral constructions. More recently, Hamiltonian quantum-simulation work has made exact plaquette operator dictionaries and economical local bases practically relevant.

This manuscript's distinctive target is narrower: the exact emergence, from $SU(N)$ fusion channels, of a signed cubical boundary operator in the fundamental C -odd one-plaquette sector. General incidence-matrix flat-band mechanisms are familiar in spectral graph theory; the nontrivial gauge-theory content is the coefficient t_N , its positivity, and its N^{-3} cancellation.

A formal novelty search should still be completed before submission.

15. Conclusion

The supplied program becomes materially larger when it is organized around a factorization rather than a chain of increasingly global claims.

On the strong-coupling side,

$$\text{group theory} \rightarrow t_N, \quad \text{cubical homology} \rightarrow \text{spec}S,$$

and the product gives an exactly flat lowest C -odd branch for every $SU(N \geq 3)$ through second order. The flatness is universal; the mobility scale is rank dependent; the N^{-3} suppression is an exact cancellation.

On the local side,

$$\text{Weyl invariant degree} \rightarrow \text{first angular correction},$$

and the $SU(3)$ cubic invariant produces a precisely quantified departure from the radial approximation.

Together these results motivate a Weyl–Hodge spectral program with two honest frontiers:

1. determine the first dynamical violation of homological flatness at $O(y^4)$; and
2. construct a rank-uniform one-plaquette limit at $\beta = N^3\tau$.

That is a bigger theory without blurring the boundary between local spectral facts, lattice band geometry, and global gauge-theory claims.

Appendix A. Exact algebra in compact form

Let

$$D_N = (N^2 - 1)(2N^2 - 1)(4N^2 - 9).$$

Then

$$\begin{aligned} A_N &= -\frac{2N^3(4N^2 - 9)}{D_N}, \\ B_N &= -\frac{4N(N^2 - 2)(2N^2 - 1)}{D_N}, \\ t_N &= B_N - A_N = \frac{2N(N^2 - 4)}{D_N}, \\ \ell_N &= A_N + B_N + \frac{2N}{N^2 - 1} = \frac{2N(5 - 3N^2)}{D_N}. \end{aligned}$$

The key numerator cancellations are

$$N^2(4N^2 - 9) - 2(N^2 - 2)(2N^2 - 1) = N^2 - 4,$$

and

$$(N^2 - 1)(4N^2 - 9) - 2(N^2 - 2)(2N^2 - 1) = 5 - 3N^2.$$

For $N = 3$,

$$\begin{aligned} A_3 &= -\frac{27}{68}, & B_3 &= -\frac{7}{18}, \\ A_3 + B_3 &= -\frac{481}{612}, & B_3 - A_3 &= \frac{5}{612}. \end{aligned}$$

Appendix B. Certificate coverage

The accompanying file `suN_homological_flat_band_certificate.py` uses only Python's standard library. It checks:

1. the channel formulas as formal rational identities in N ;
2. $t_2 = 0, t_N > 0$ for $3 \leq N \leq 128$, and the $SU(3)$ regression values;
3. the leading coefficients

$$N^3 t_N \rightarrow \frac{1}{4}, \quad N^3 \ell_N \rightarrow -\frac{3}{4}, \quad 12N^3 t_N \rightarrow 3;$$

4. $\det \tilde{N} = 0$ as a Laurent-polynomial identity;
5. the cross-product Gram identity and flat kernel;
6. exact rational ranks of the periodic ∂_2 matrices at $L = 2, 3$.

It intentionally does not certify the source-only $SU(3)$ third-order number.

Appendix C. Source-provenance map

Source file	Role in the enlarged manuscript
Non-Uniformity in N and Formal t Hooft Rescaling.docx	$\beta \gg N^3$ and formal $\beta = N^3 \tau$ scaling
SU(3) Weyl Geometry and Local Hamiltonian.docx	Cartan invariants, discriminant, scaled operator
Wilson Hard-Defect Peierls Bound.docx	conditional global layer only
From Local Class Gaps to Global Wilson Geometry.docx	logical separation of local and global claims
Rooted Projected-Capacity Source Stability.docx	projected-capacity status and limitations
Compact Character-Basis Numerical Audit.docx	numerical local diagnostics
Finite Leakage Matrix and Radial Tail Obstruction.docx	finite truncation diagnostic; not a polymer theorem
Theorem Ledger and Open Analytic Targets.docx	status discipline
SU(3) Three-Term Local Gap.docx	exact local gap theorem
duplicate SU(3) Weyl Geometry... file	byte-identical duplicate; not double-counted
Logical Status and Theorem Map (1).docx	corrected fixed-rank formulas and status map
Numerical Projected-Capacity Audits.docx	numerical global diagnostics
Finite Wick Certificate for the SU(N) C-odd Local Gap.docx	fixed-rank odd-sector certificate
SU(3) Coefficient Tables and Matrix Elements.docx	exact finite basis and matrix elements
Fixed-Rank SU(N) Local Class Gaps.docx	fixed-rank $C_{\pm}$ theorems
Sun report .md	project-level synthesis
flat band full derivations.tex	SU(3) strong-coupling band, topology, third-order source claim

References

1. J. Kogut and L. Susskind, “Hamiltonian formulation of Wilson’s lattice gauge theories,” *Physical Review D* **11**, 395 (1975). doi:10.1103/PhysRevD.11.395
2. A. M. Perelomov, “A simple construction of basic polynomials invariant under the Weyl group of the simple finite-dimensional complex Lie algebra,” *Communications in Mathematics* **28**, 301–305 (2020). doi:10.2478/cm-2020-0024
3. W. Garcia Fuertes, M. Lorente, and A. Perelomov, “An elementary construction of lowering and raising operators for the trigonometric Calogero–Sutherland model,” *Journal of Physics A* **34**, 10963–10973 (2001). arXiv:math-ph/0110038
4. R. G. Jha, “Finite N unitary matrix model” (2020). arXiv:2003.00341
5. P. Balaji *et al.*, “Quantum Circuits for SU(3) Lattice Gauge Theory” (2025). arXiv:2503.08866
6. T. Jakobs *et al.*, “A Comprehensive Stress Test of Truncated Hilbert Space Bases against Green’s function Monte Carlo in U(1) Lattice Gauge Theory” (2025). arXiv:2510.27611
7. V. Kazakov and Z. Zheng, “Bootstrap for Finite N Lattice Yang–Mills Theory” (2024). arXiv:2404.16925